

Public Economics

Lecture 10 · Political Economy

Shushanik Margaryan

Winter Term

WU Vienna University of Economics and Business

Today's Questions

- Public goods theory tells us *how much* government should provide — but who actually decides that amount, and how?
- Does majority voting reliably aggregate everyone's preferences into a sensible collective choice?
- If government itself is run by self-interested politicians, bureaucrats, and lobbyists, can we still trust it to fix market failures?

Based on

Gruber, J. *Public Finance and Public Policy*, Ch. 9 (Political Economy).

9.1 Direct Democracy

9.2 The Median Voter Theorem

9.3 Representative Democracy

9.4 Is Government Efficient?

9.1 Direct Democracy

From “How Much” to “Who Decides”

The Samuelson rule tells us the *efficient* quantity of a public good. It says nothing about **who determines** that quantity in practice.

- Real preferences for public goods are private information — nobody has an incentive to reveal them truthfully if doing so raises their tax bill.
- Democracies use some form of **voting** to aggregate millions of individual preferences into one collective choice.

The central question of this lecture

Does the political process that decides *how much* government to have itself work efficiently — or can *government failure* be just as real as market failure?

Unanimity and the Lindahl Solution

Lindahl pricing

Charge each person a personalized “tax price” per unit of the public good, set so that, at that price, everyone *independently* demands exactly the same efficient quantity — and unanimously agrees to it.

Why it fails in practice:

- Nobody has an incentive to reveal their true valuation — understating it lowers your personalized tax price (the same preference-revelation problem from public-goods theory).
- Reaching genuine unanimity among millions of people is essentially impossible — a single holdout can block everything.

In practice, societies use **majority voting** instead of unanimity — trading away first-best efficiency for a decision rule that can actually produce a decision.

9.2 The Median Voter Theorem

Single-Peaked Preferences and the Median Voter Theorem

Single-peaked preferences

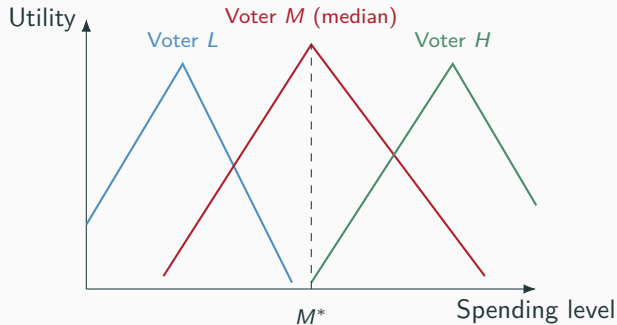
As spending moves away from a voter's ideal point in *either* direction, that voter's utility falls monotonically.

Median voter theorem (Black, 1948)

If all voters have single-peaked preferences over a single policy dimension, majority voting selects the **median voter's** preferred outcome — and that outcome is stable: no other option can beat it in a head-to-head majority vote.

This is a remarkably strong result: it doesn't matter how extreme the preferences at either tail are — only the voter exactly in the middle determines the outcome.

Diagram: The Median Voter Wins



With single-peaked preferences, the median voter's ideal point M^* beats every other proposal in pairwise majority voting: any move toward L 's peak loses M 's and H 's votes; any move toward H 's peak loses M 's and L 's votes.

When the Median Voter Theorem Breaks Down

The theorem requires single-peaked preferences over a **single** dimension. It can fail when:

- Choices are **multi-dimensional** (e.g., voting simultaneously on tax rates *and* spending priorities) — “single-peaked” may not even be well defined.
- Some voters have **double-peaked** preferences (e.g., someone who wants either very low or very high spending, but not a moderate compromise).

The Condorcet paradox

Even with perfectly rational, transitive individual preferences, **majority voting itself can be intransitive.**

Illustration: The Condorcet Paradox

Three voters, three options, each individually rational:

	1st choice	2nd choice	3rd choice
Voter 1	A	B	C
Voter 2	B	C	A
Voter 3	C	A	B

Pairwise majority votes: A beats B (2–1) B beats C (2–1) C beats A (2–1).

A voting cycle

$A \succ B \succ C \succ A$ — majority rule is **intransitive**, even though every individual voter is perfectly consistent. Whoever controls the voting *agenda* (which pairs get voted on, in what order) can determine the winner.

Arrow's Impossibility Theorem

Kenneth Arrow (1951) asked a deeper question: is there *any* voting rule that avoids problems like the Condorcet paradox while satisfying basic fairness conditions?

- No dictator.
- Every individual preference ordering is allowed.
- If everyone prefers A to B , so does society.
- The social ranking of A vs. B doesn't depend on preferences over some unrelated C .

Arrow's result

No voting rule can satisfy **all** of these simultaneously, once there are three or more options. There is no perfect, universally fair way to aggregate individual preferences into a social ranking.

9.3 Representative Democracy

From Direct to Representative Democracy

Direct voting on every policy question is impractical at national scale — societies instead elect **representatives** who vote on their behalf.

- Representatives face their *own* incentives (re-election, party loyalty, campaign funding) — not necessarily their constituents' median preference.
- Representatives routinely vote on **bundles** of issues at once, opening the door to vote-trading.

Logrolling and Pork-Barrel Spending

Logrolling

Trading votes across issues: “I’ll support your project if you support mine.”

Example: two projects each cost \$120 (split evenly across all districts) but benefit only *one* district by \$100 each — neither passes on its own merits. But if the two district representatives agree to support *both* projects, each gets a project that benefits their own district by \$100 at a shared cost of \$120 split broadly — and a majority coalition can form to pass both, even though neither is efficient in isolation.

Legislative “universalism”

When this logic generalizes across many districts, legislatures can end up approving a long list of narrowly-targeted, individually inefficient projects — classic “pork-barrel” spending — because the costs are spread thin across everyone while the benefits are concentrated.

Olson's logic of collective action (1965)

A policy with **concentrated benefits** to a small group and **diffuse costs** spread across everyone else is far more likely to pass than the reverse — even when the diffuse costs exceed the concentrated benefits.

- The small group has a strong individual incentive to organize, lobby, and monitor the issue.
- Each individual bearing the diffuse cost has almost no incentive to fight back — their own share of the cost is tiny (a free-rider problem, mirror image of underprovision of public goods).

This asymmetry is a standard explanation for why narrow industry protections and subsidies persist even when they plainly fail a cost-benefit test.

Regulatory capture (Stigler, 1971)

Regulatory agencies, created to police an industry in the public interest, tend over time to be captured by — and act in the interest of — the very industry they regulate.

- The regulated industry has concentrated stakes and deep expertise, and lobbies intensively.
- The diffuse public — the intended beneficiary of regulation — has little organized voice in the process.
- Revolving-door career patterns between regulators and industry can reinforce the alignment of interests.

9.4 Is Government Efficient?

The Leviathan Model of Government

Leviathan model (Brennan & Buchanan, 1980)

Instead of a benevolent social planner maximizing welfare, treat government as a **revenue-maximizing monopolist** that expands its own size and budget as far as voters and institutions will allow.

- Implication: constitutional or statutory limits on tax rates and debt aren't just red tape — they're a deliberate check on a self-interested state.
- Competition *between* jurisdictions (people and firms can move) is one market-like discipline on a Leviathan government — akin to Tiebout competition.

Bureaucracy: The Niskanen Model

Niskanen model (1971)

Bureaucrats are assumed to maximize their agency's **budget** (which brings prestige, staff, and influence) rather than social welfare.

- A budget-maximizing bureaucracy tends to **overprovide** its service relative to the efficient level — the mirror image of the private-sector *underprovision* of public goods.
- Because legislators often can't easily verify the agency's true cost structure, the agency can credibly claim it needs a larger budget than efficiency requires.

Not every empirical study finds bureaucratic overprovision — but the model is a standard benchmark for thinking about principal–agent problems between voters, legislators, and the agencies that actually implement policy.

Conclusion: Weighing Market Failure Against Government Failure

Market failure (externalities, public goods, imperfect information) is a reason government **could** improve on the market outcome — **not** a guarantee that it **will**.

- Voting mechanisms can be manipulated, cycle, or simply fail to aggregate preferences well (median voter breakdown, Condorcet paradox, Arrow's theorem).
- Representatives, bureaucrats, and regulators all have their own objectives, which can diverge from social welfare (logrolling, capture, budget-maximization).

The right comparison

The relevant policy question is never “markets vs. perfect government” — it is **this market failure vs. this specific government intervention**, warts included on both sides.

Sources

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Thank you!

Questions and discussion